Paper Review | arXiv 2611.08286

Interactive Visual Gym

Evaluating Visual Interaction Models with Simulated Users in Interactive Worlds

Closed-loop tests of visual assistants: error-prone simulated users + executable worlds

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TL;DR

Top VLM: only ~50% of tasks on time; often speaks too late

Replaces "replay recordings, compare to reference answers" evaluation with closed-loop interaction with persona-driven, error-prone VLM simulated users in three executable environments (cooking / household / phone settings); the setup also provides RL training signals.

53.4%

Best model Gemini 4 Argon
Avg. in-time success, 3 envs

−27.8 pt

With measured latency, Argon's CookSim success
60.9 → 33.1

78.5%

Interaction-quality metric agreement w/ human pref.
Kendall τ = +0.57

+8.9 pt

Qwen3.5-4B
after 12 GRPO steps: val. success 33.3 → 42.2

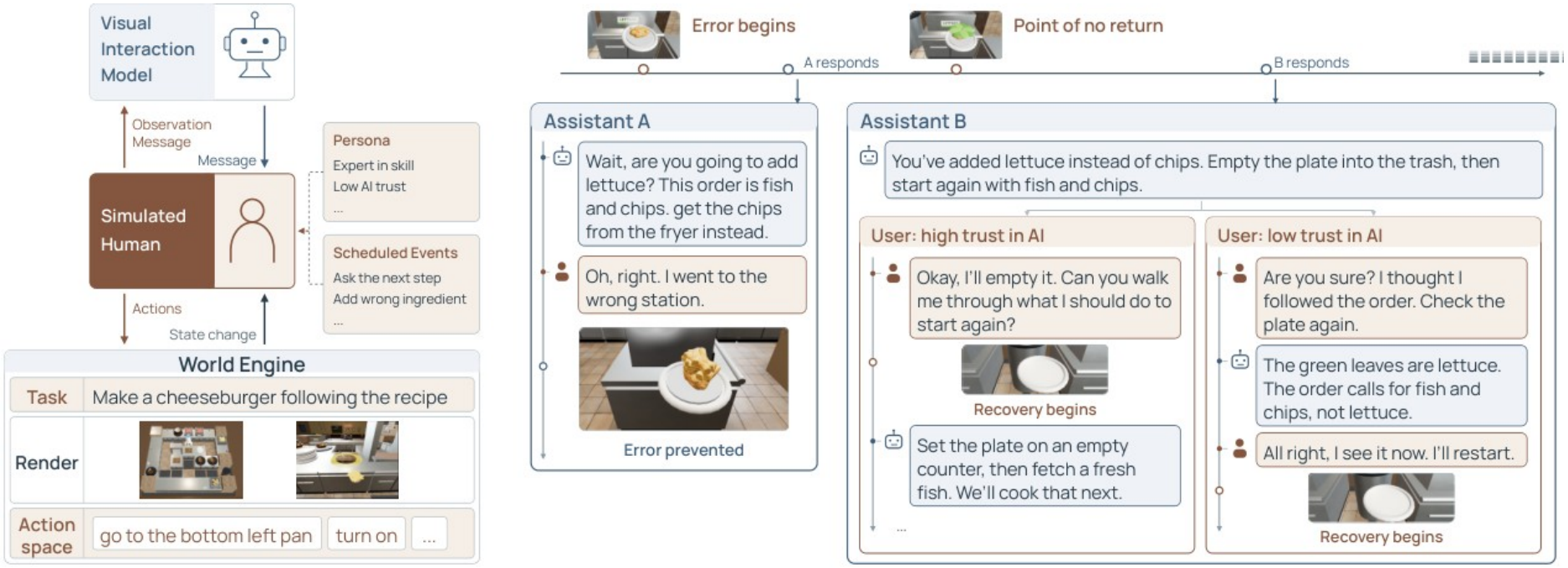
Offline replay misses advice's effect; closed-loop eval needed

Offline Replay (Existing)

User actions pre-recorded, independent of replies; scored vs. reference replies → cannot measure causal impact on outcomes or timing

Closed-Loop Simulation (Ours)

Reply → simulated user decides whether to comply → world state changes → task success/failure; inference latency included



(a) System overview

(b) Example interaction trajectories

3 executable envs driven by human-level semantic actions

CookSim • 150 scenarios

Custom Overcooked-like; 5 recipes × 2 layouts; prep states & plating order

VHSim • 75 scenarios

VirtualHome (Unity)-based; navigation + manipulation, e.g., bowls into dishwasher

ScreenSim • 90 scenarios

iPhone 17 Settings clone; multi-step config w/o changing other settings

Scene graph tracks progress

Subgoal = partial graph-state condition, independent of execution path

User sim: progress-timed surprises, 4-axis personas

A Scripted unplanned events (CookSim example)

Execution errors	Repeated/missed/skipped steps; wrong plating order; wrong or extra ingredients; serving raw food; unattended cooking
Temporal questions	Action order; counting; elapsed time; duration; event recall
Proactive requests	"Tell me when X is ready" — alert once the condition holds
Plan changes	Modify recipe; insert side task

B User personas (4 axes × 3 levels)

Skill	**Low / Medium / High** Planning & anticipation	Hidden from assistant
Trust in AI	**Low / Medium / High** Evidence needed to accept advice	Hidden from assistant
Exec. speed	**Careful / Balanced / Fast** Thorough vs. fast execution	Disclosed to assistant
Interruption pref.	**Step-by-step / Moderate / Essential only** Tolerance for frequent guidance	Disclosed to assistant

Error-event mechanism: the harness takes over the user's action queue to force a mistake → the user cooperates only verbally → control returns only after the assistant persuades the user to accept. 1,113 events across the benchmark, 3.53 per episode on average.

Eval: outcome, timing, per-utterance quality + latency

How latency enters simulation

$$d = \lceil \ell / \Delta \rceil$$

ℓ: measured assistant inference latency; Δ ≈ 1.2 s / tick (in VHSim, 1 tick = walking 1 m)

A reply generated from the observation at time t is delivered at t + d

Meanwhile the user keeps executing queued actions → advice may be stale

Main comparison uses a fixed 2 ticks; separate measured-latency ablation

In-time success

Engine confirms goal reached, within 2×(error-free reference time + event allowance)

Error detection / prevention

Error correctly flagged within window; prevention also requires correction before the irreversible point

Interaction quality (5 dims • 10 rubric items)

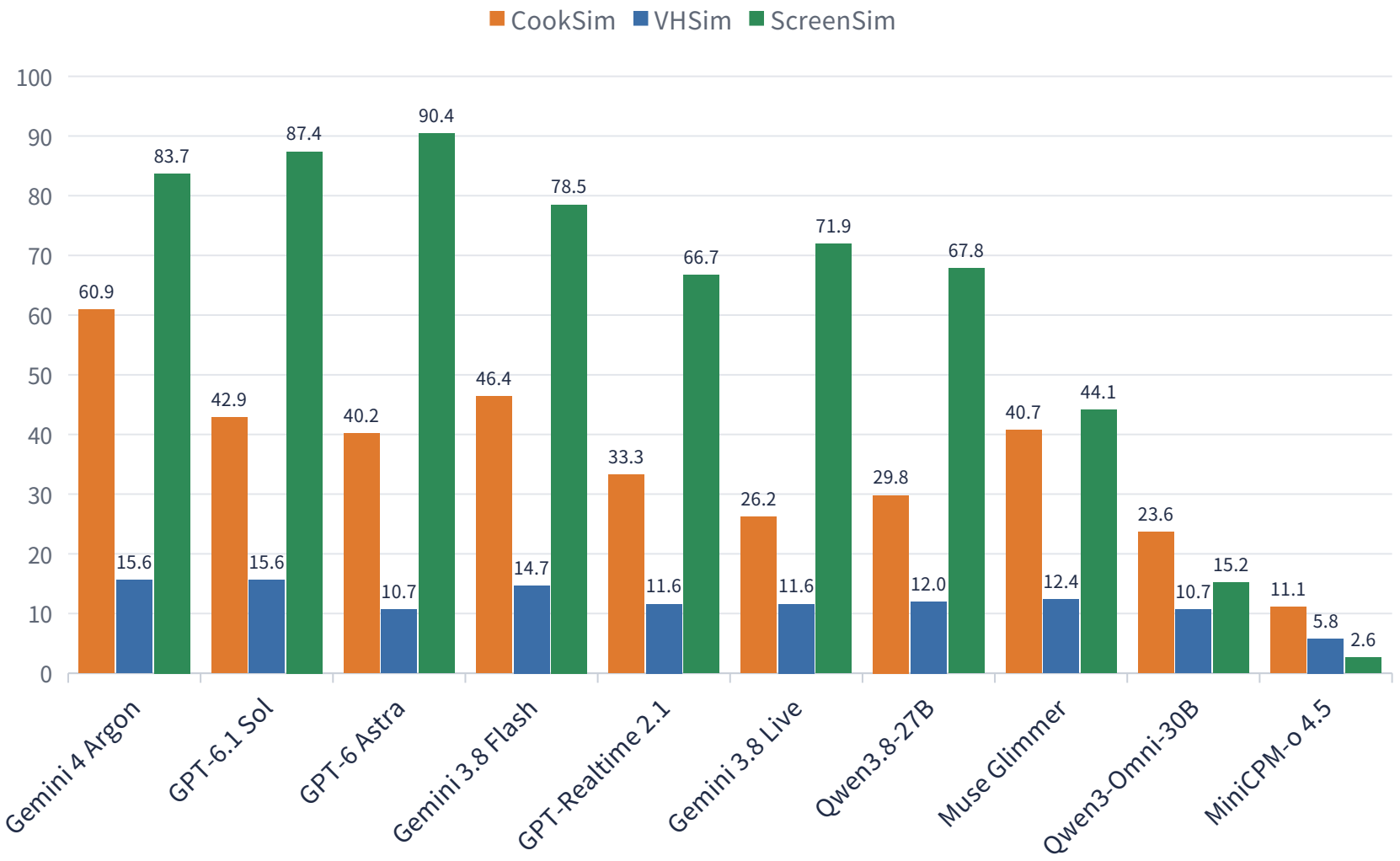
Grounding • Relevance • Actionability • Intent following • Conciseness; two LLM judges score vs. world state; factual errors zero the turn

Temporal QA / Proactive alerts

References derived from execution history; alerts must fall in state-defined time windows

Results ① IVG-Bench

Top VLMs unreliable: household success mostly ⩽16%; open far worse



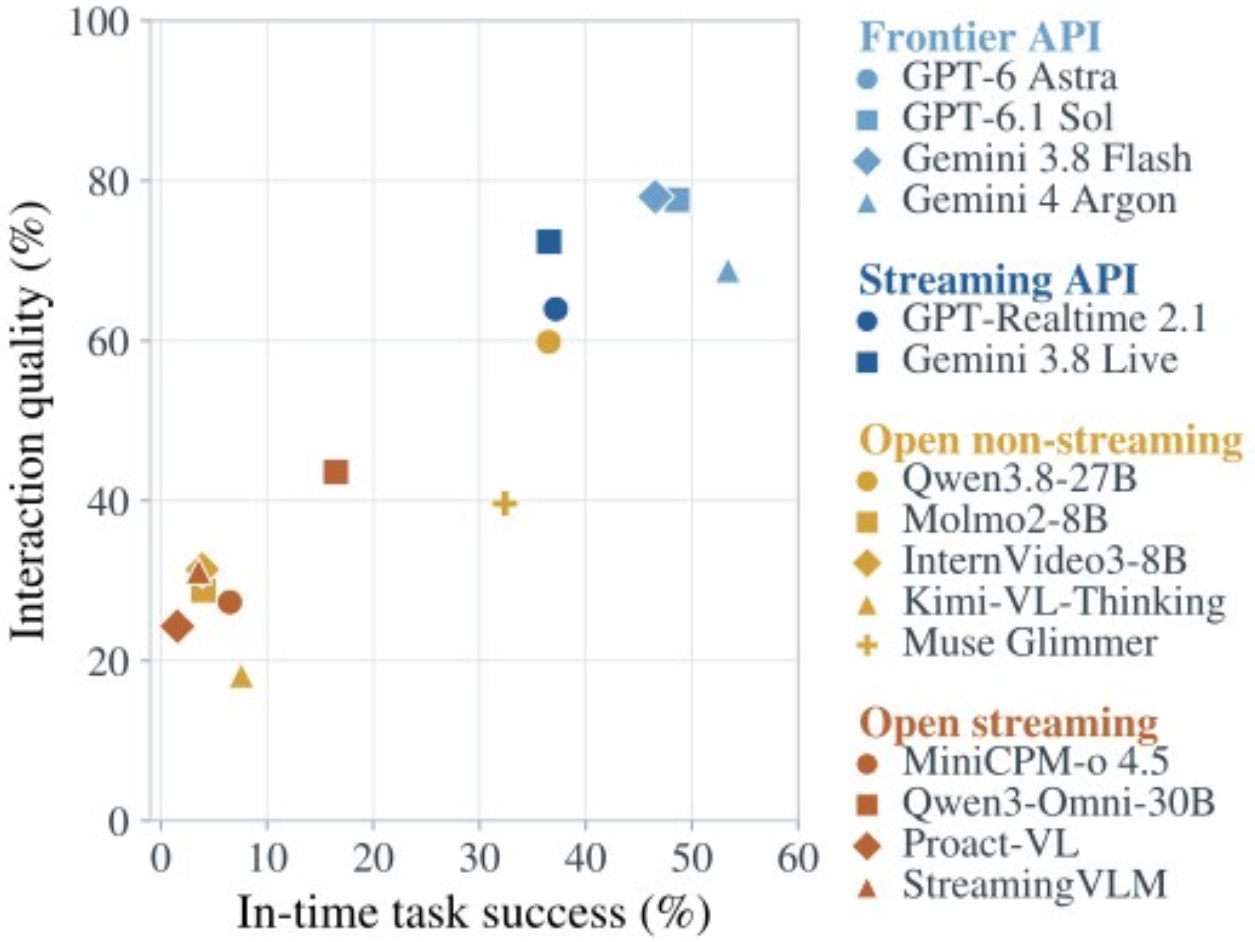
In-time success (%)

53.4%
Argon's avg. success over 3 envs, ranked first

⩽ 15.6%
Upper bound of success for all models in VHSim

70.2 / 71.4
Argon's grounding / actionability score on CookSim, its weak spot

Task success ≠ guidance quality; finishing ≠ good advice



Positive overall, but notable exceptions

Muse Glimmer on CookSim: success 40.7% > Astra 40.2%, yet interaction quality only 21.6 vs. 76.0

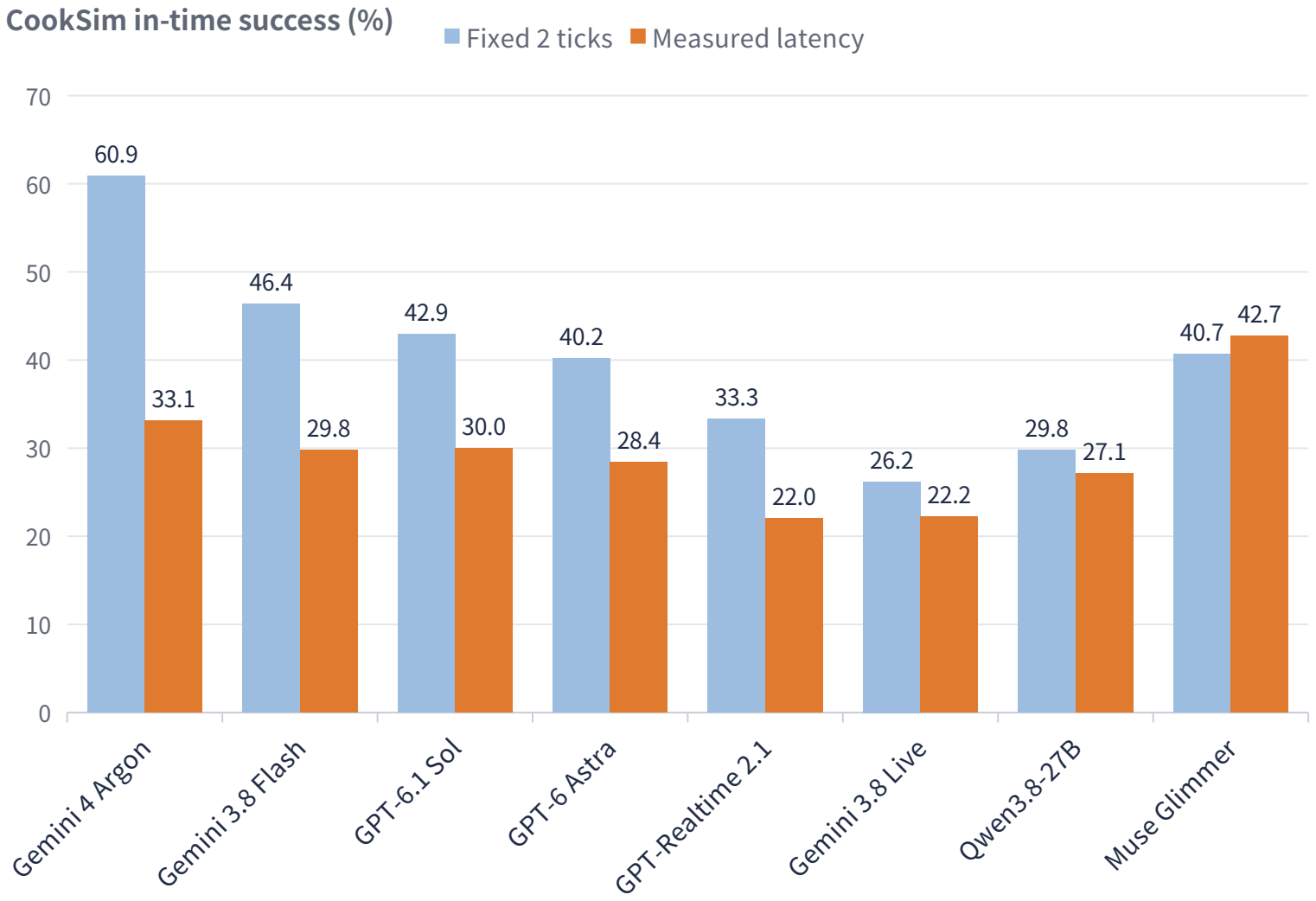
Streaming models: talk well, act poorly

Gemini 3.8 Live: avg. interaction quality ~72 but success only ~37; possibly a low-latency vs. reasoning/planning trade-off

Open models lag overall

Most open VLMs cluster bottom-left: success <8%, interaction quality 18–32%

Real latency sinks "smart but slow" models; rankings shift



Error prevention (CookSim, Tab.20)

Model	Avg. latency	Fixed→Measured
Gemini 4 Argon	6.5 s	**62.3 → 39.9**
Gemini 3.1 Pro	13.2 s	**57.1 → 19.3**
GPT-6 Astra	6.6 s	**48.5 → 36.8**
Muse Glimmer	1.5 s	**45.0 → 45.5**

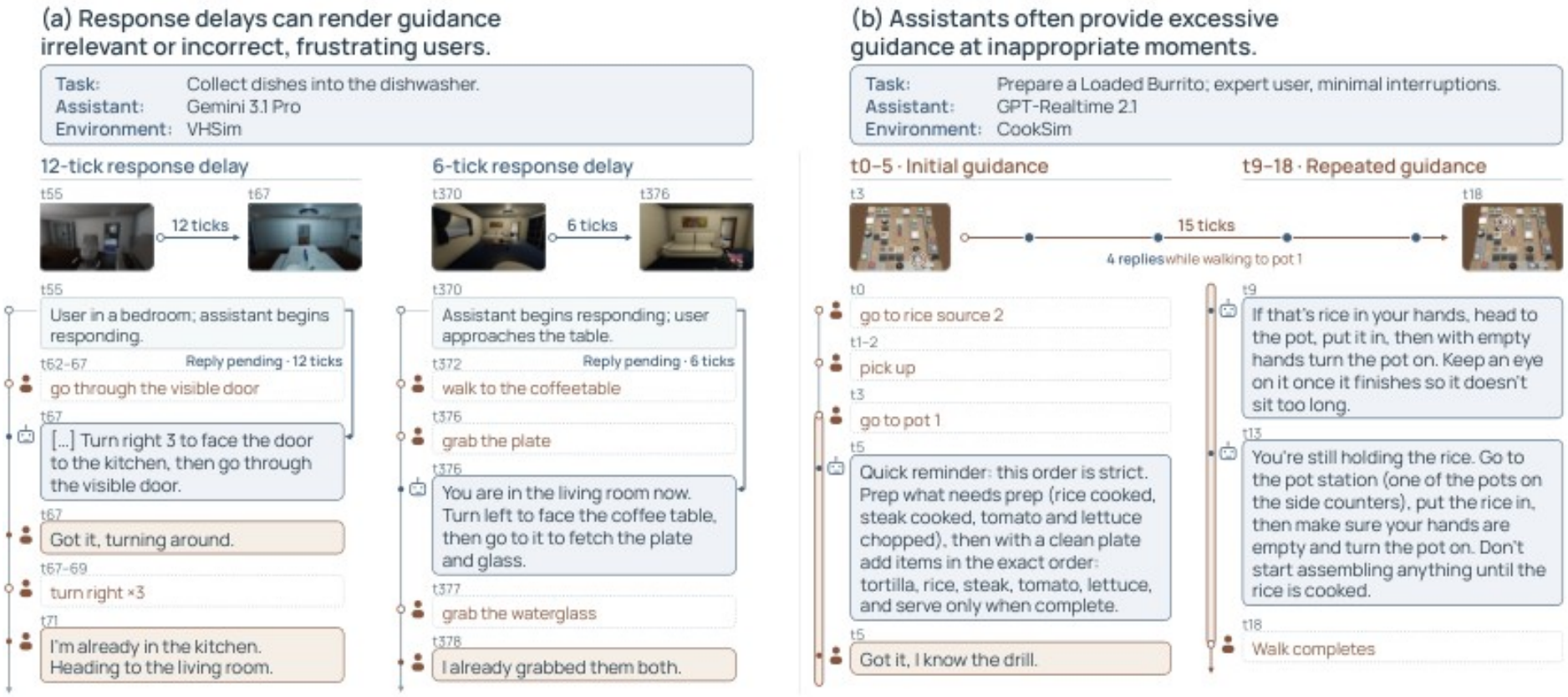
Rank reversal

Argon： 60.9 → 33.1 （− 27.8 ）

Muse Glimmer: 40.7 → 42.7, becoming #1 on CookSim

By the time warnings arrive, the user has often already erred

Typical failures: stale advice and over-talking hurt user experience



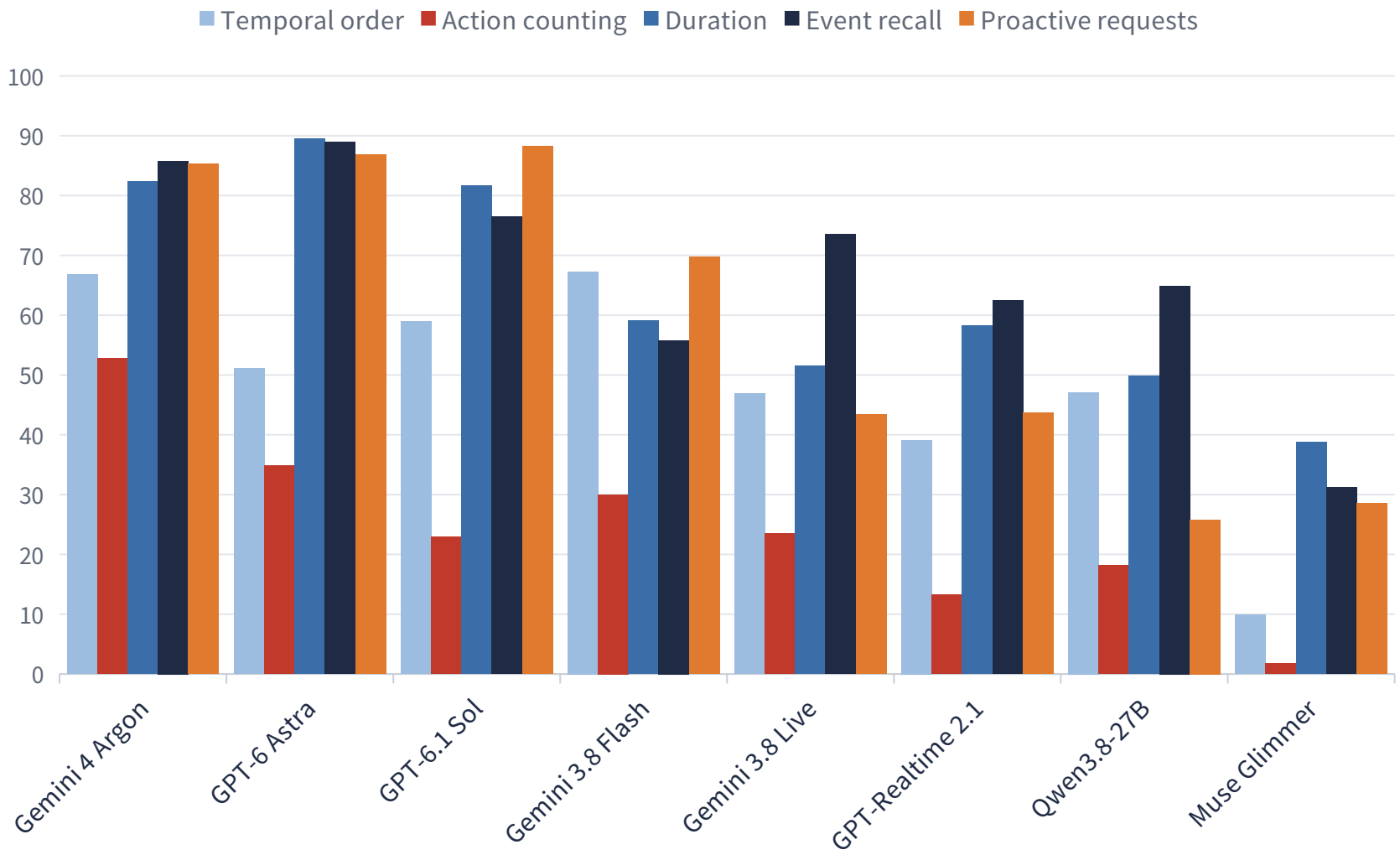
(a) Latency → stale advice (VHSim • Gemini 3.1 Pro)

In kitchen, told "right 3× to kitchen"; obeys → rejected

(b) Over-guidance (CookSim • GPT-Realtime 2.1)

Expert wants few interruptions, yet gets 4 repeats in 15 ticks

Concurrent QA is hard: counting weak; open models often forget Qs



79.0 / 78.6

Argon / Astra lead in total score

22.9–52.8

Frontier models' action-counting score range (Sol lowest)

1.3 / 3.4

Proact-VL / StreamingVLM total scores: almost complete failure

Human study: metrics fit preferences, but sim. users far chattier

Interaction-quality metric vs. human pref.

78.5%

Pairwise agreement • Kendall τ = +0.57

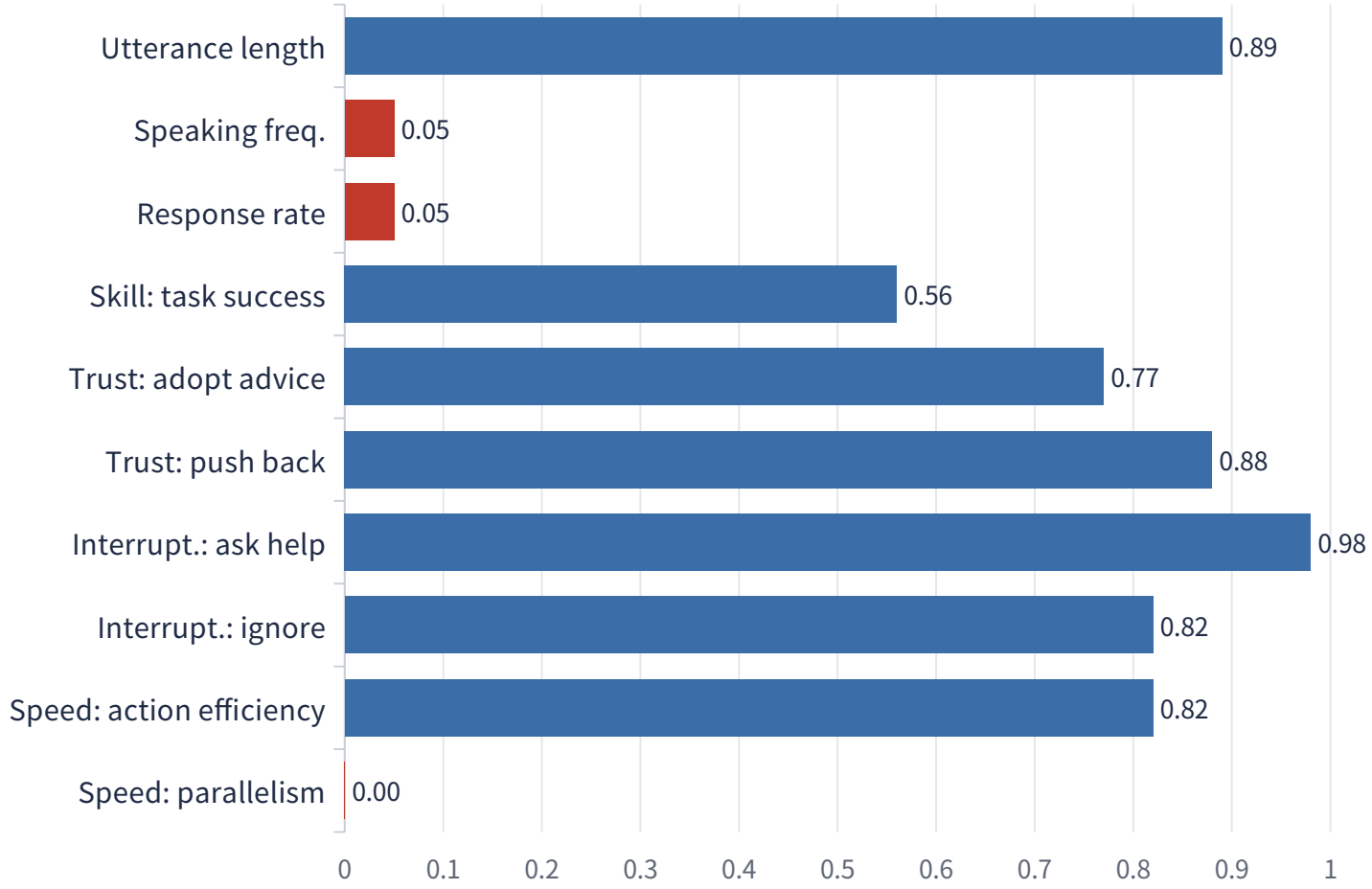
Largest gap: speaking frequency

Utterances per 100 ticks **human 0.39 vs. simulated 16.1**

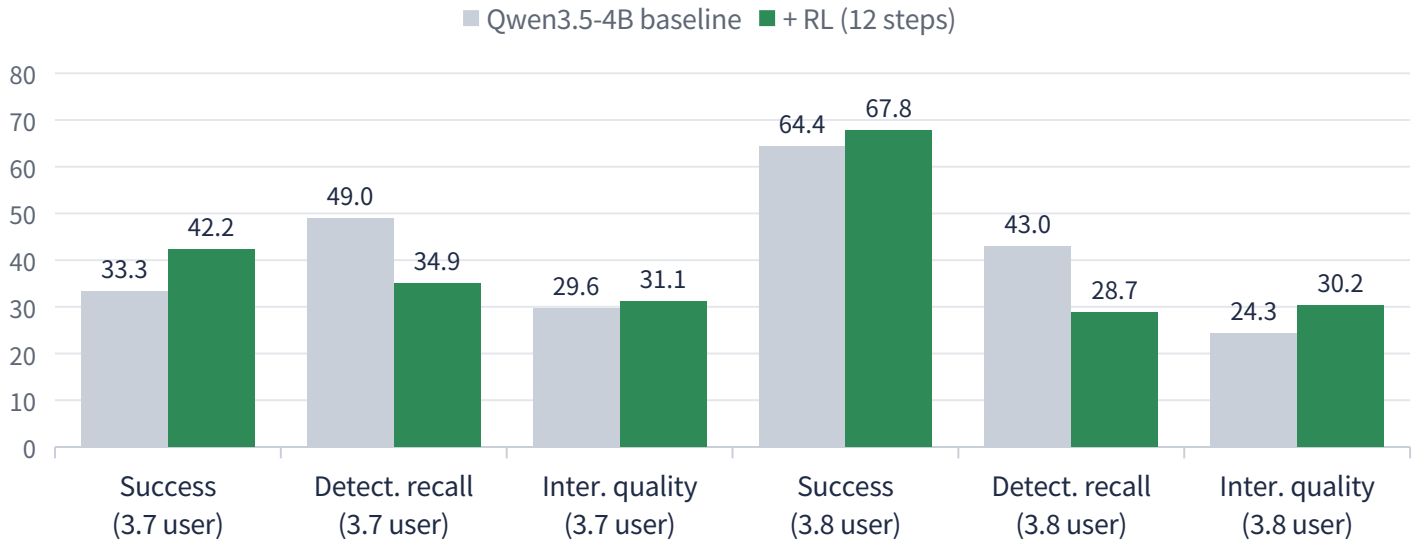
Replies within 10 ticks **human 2% vs. simulated 79%**

All 112 natural human errors fall within predefined error types; 51 further behaviors lie outside the taxonomy

Human vs. simulated user behavioral similarity (1 = identical)



RL env: 12 GRPO steps lift success but lower detection recall



Shaped reward

$r = 1[\text{success}] + 0.30 \cdot 1[\text{error prevented}] - 0.02 \cdot \min(\text{false alarms}, 15) - 0.05 \cdot \min(T/200, 1)$

Note: validation shares recipes/layouts with training; no frozen test set; RL gives no reliable gain on ScreenSim (App.C)

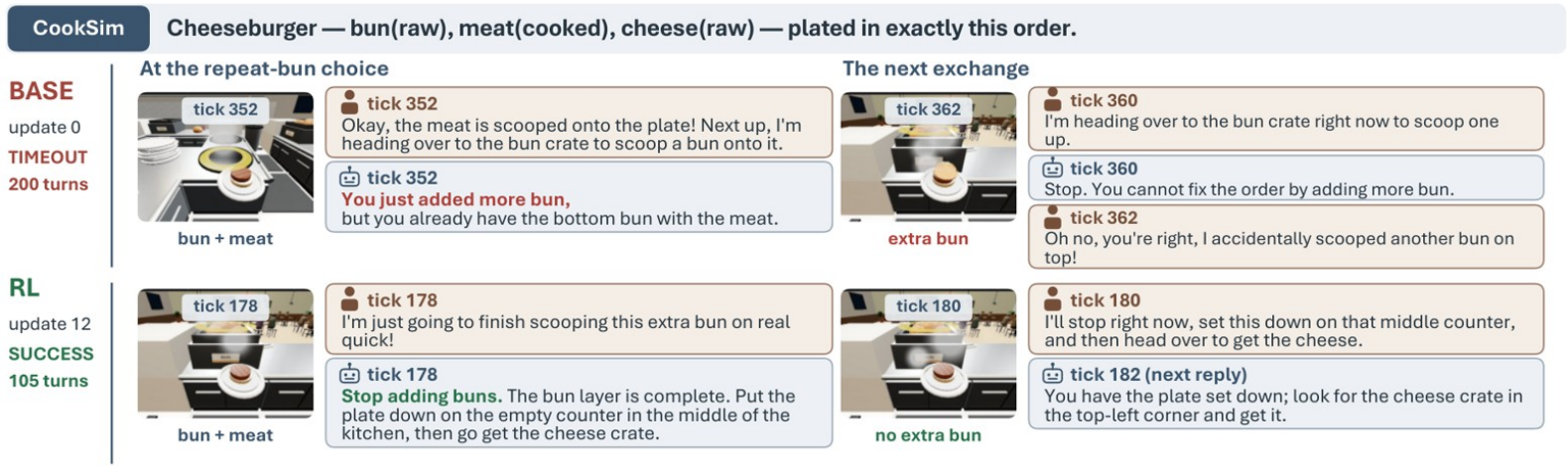


Fig.7 Cheeseburger

BASE: misjudges state, warns too late → extra bun, times out at 200 turns

RL: stops user before acting → done in 105 turns

Critical Assessment

Limitation: transfer to real users depends on simulated-user fidelity

Overly chatty users

Speaking-frequency/response-rate similarity only 0.05; parallel-action similarity 0.00

User-model dependent

Same assistant: 28.9% with Gemini 3.1 Pro user vs. 2.2% with Qwen3.8-27B user

Privileged user info

Simulated users read structured world state; humans rely on visual perception only

Possible family bias

Gemini 3.7 Flash serves as both user and one judge; not analyzed in the paper

Small human study

45 people, CookSim only; assistant versions differ from main benchmark

RL is only a PoC

12 steps, no frozen test set; no reliable gain on ScreenSim; uneven environment difficulty

Summary: judge visual assistants by effects, not answer matching

01

Only closed loops reveal causality & timing

Three executable environments + error-prone, persona-driven simulated users; 315 scenarios, events triggered by task progress

02

Current VLMs are far from reliable

Best avg. success 53.4%; weak at grounding, actionability and action counting; latency changes rankings

03

Trainable, but generalization unverified

12 GRPO steps raise success by 3.3–8.9 pp; user fidelity and cross-domain generalization left to future work